DESCENT

MIDTERM 08C: MOTORIZED SOUND MACHINE PROTOTYPE

DES INV 22: FALL 2022

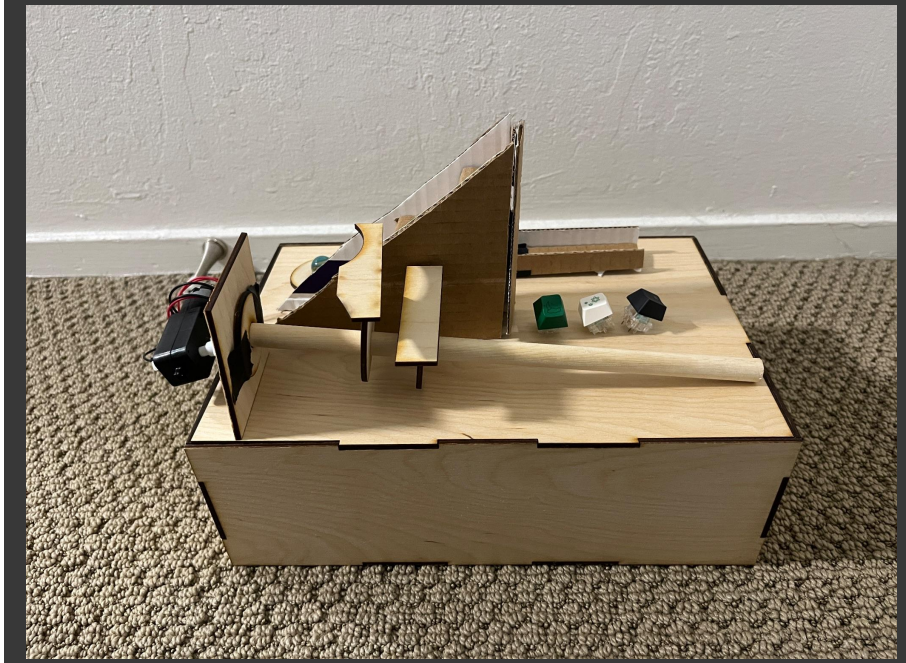
11/2/2022

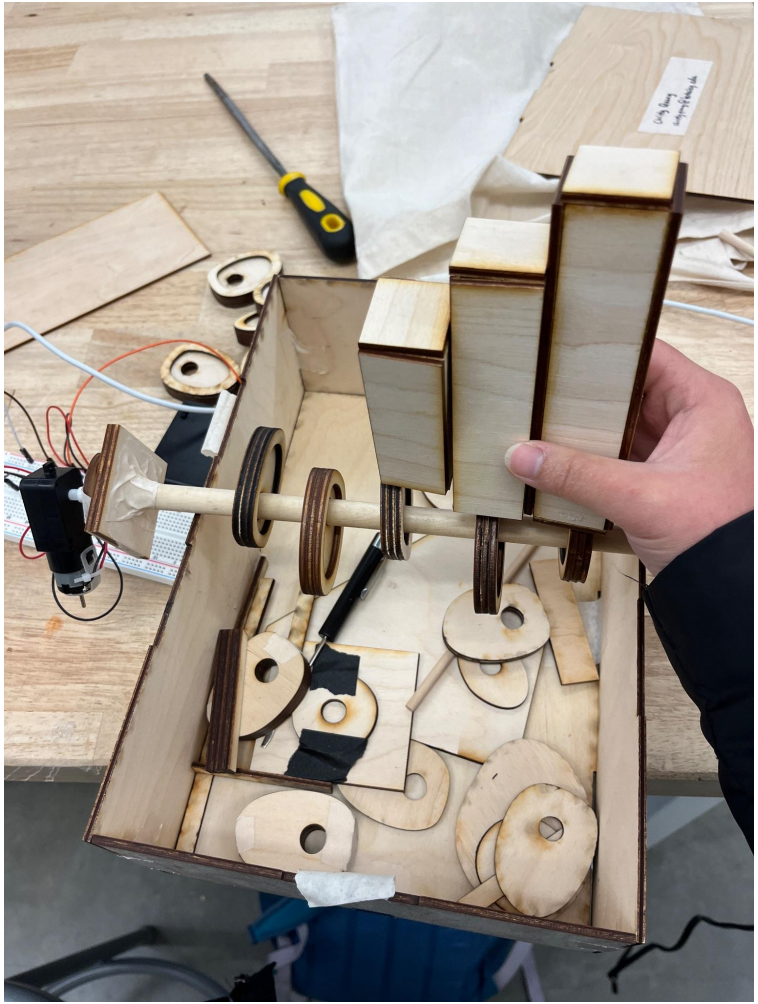
Christy Quang

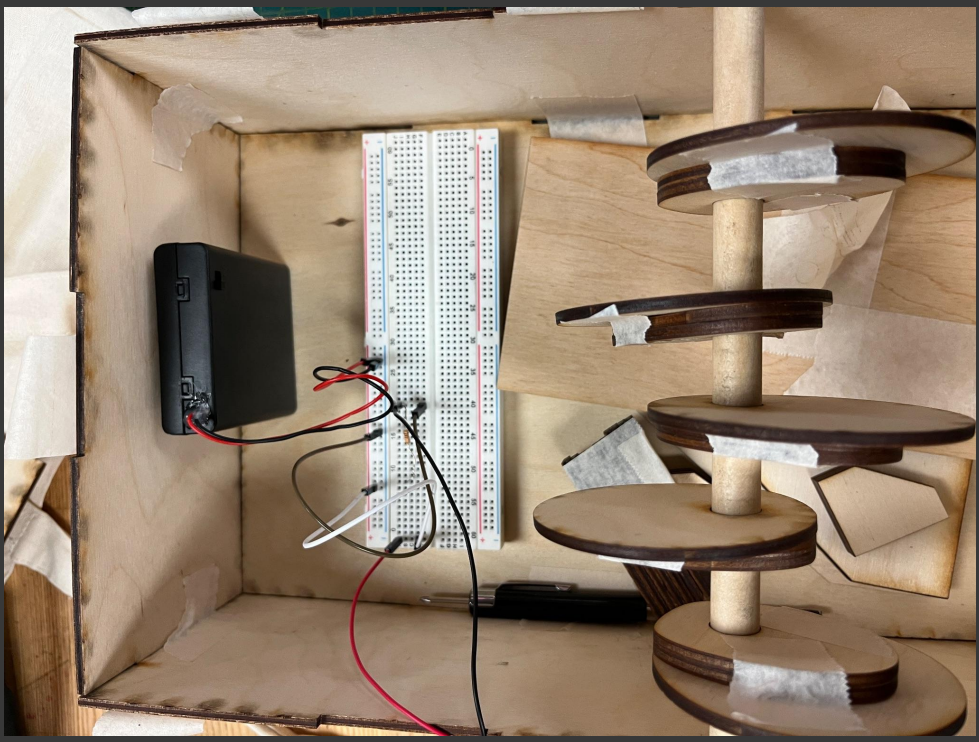
DESCRIPTION

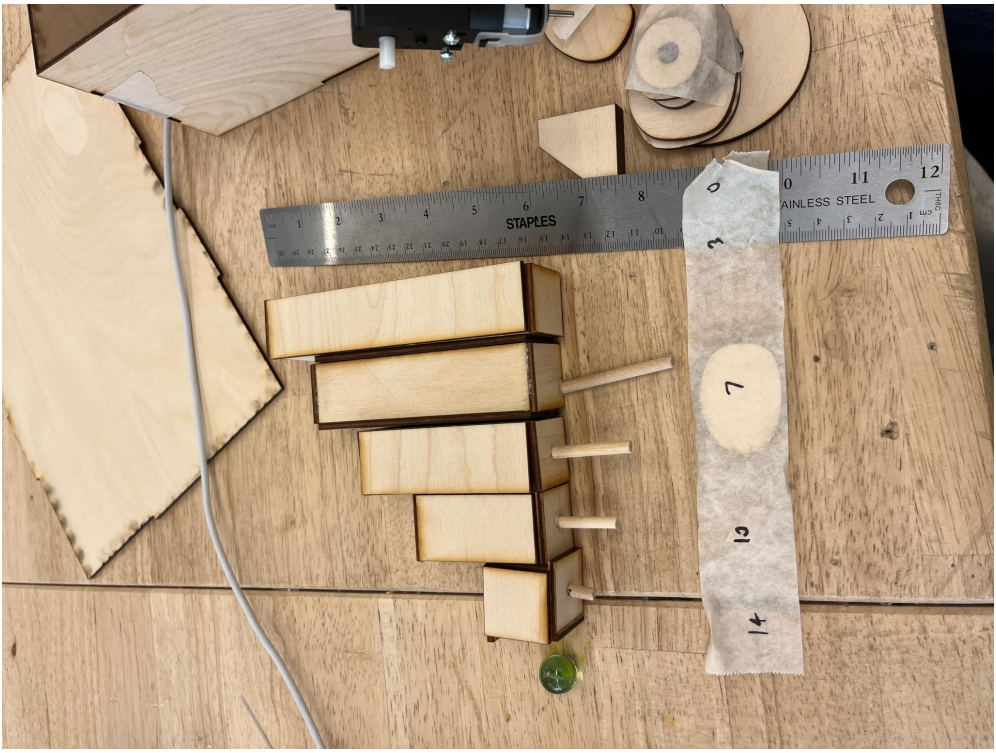
An automatic staircase to make marbles ascend, fall onto keyboard switches to create a musical rhythm, then repeat the process.

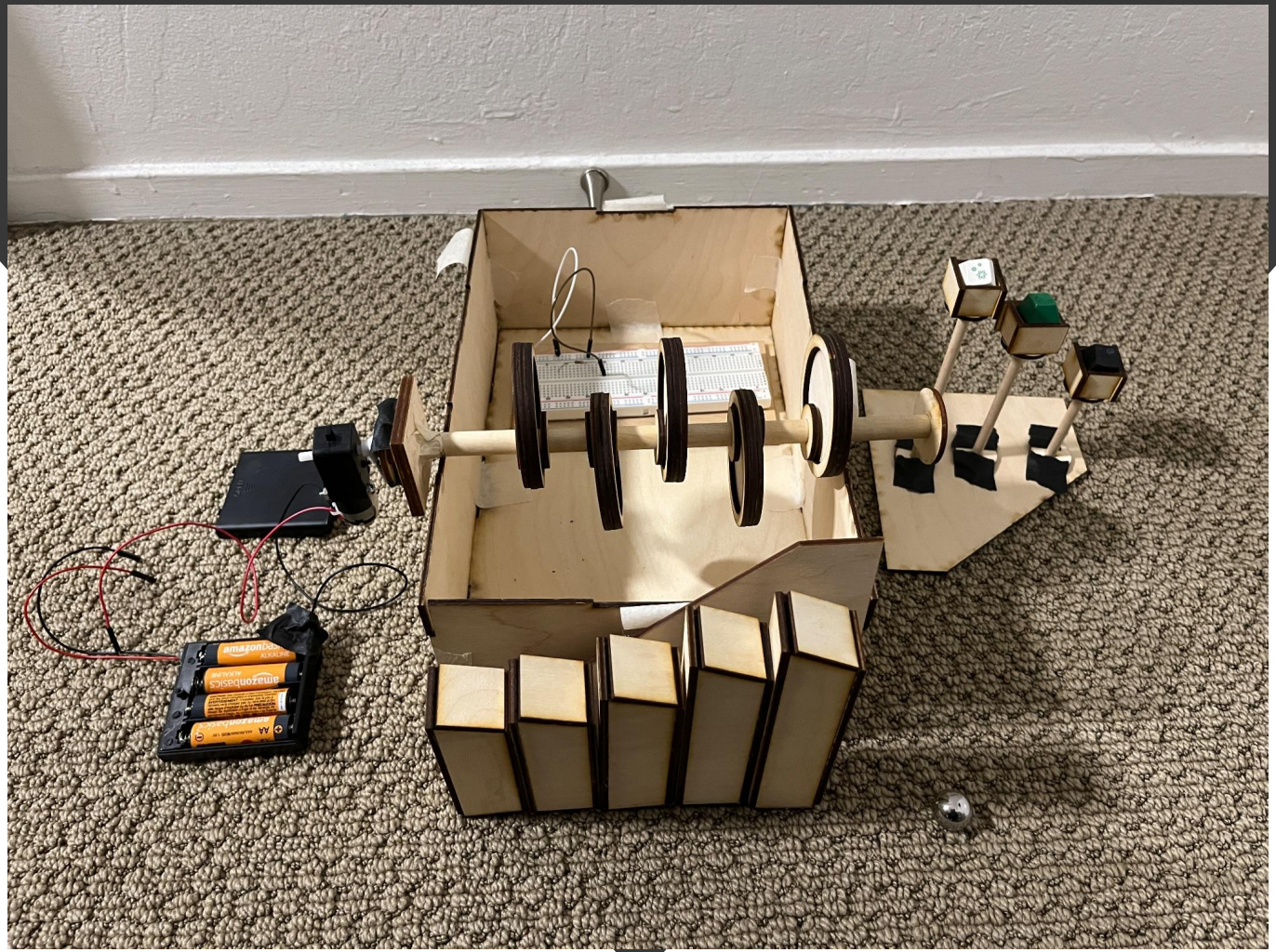
Video Link: https://youtu.be/VoYL5kdZzxg








STAPLES
STAINLESS STEEL


amazonbasics
amazonbasics

PROCESS

The automatic staircase functionality was a very difficult and unprecedented obstacle. The initial plan for the cams, staircase design, motor movement, and switch holders were all flawed and had to be changed a day before the due date. The cams were the trickiest to deal with since there was minimal understanding of how the size of the cams would affect the movement height-wise of the stairs.

There were three iterations of creating the cams and stairs. The first one was when the cams were 5 plywood layers thick and each stair was 7 layers thick, which was too heavy. The second iteration was when the cams were 4 layers thick but this was still too heavy and the cams weren't large enough or the right size to get the stairs to move correctly. The current iteration is after changing the shape of the cams and staircase design, creating a slope for the marble to roll down while also having the cams be circular to have a uniform ascend.

Building the circuit for the gear motor was smoother than expected with the help of Berk but the gear motor was too slow and weak for the stairs created. After removing a resistor, the motor was too fast but there isn't a way to adjust the speed of the motor besides programming an Arduino or adding a smaller and larger gear to alter the speed of the dowel rotation. Having a resistor would slow down the motor but this would reduce the power of the motor, causing it to be unable to withstand the weight of the stairs.

Bill of Materials

- 1 - ½” Wooden dowel
- 1 - Right Angle Gear Motor
- Battery box
- Breadboard
- 1- 20 ohm Resistor
- 3 - Cherry MX Blue Mechanical keyboard switches & keycaps
- Birch plywood
- Skinny dowels
- Acrylic plastic
- Wood glue
- Handsaw
- Sandpaper
- Paste wax
- Laser cutter
- X-acto knife
- Black Gaffers Tape

Motorized mechanism

My motorized mechanism involves using a right angle motor attached to a dowel with cams in opposite directions, allowing the stairs to move marbles up each step in a fluid motion. I thought that the motor was too weak since very little friction would cause the motor to stop moving but this was due to my high level resistor.

However, without the resistor, the motor has too much power so it rotates the dowel too fast for the marbles. To combat this speed, I want to still keep the power which I can redistribute by adding a large and small motor to change the speed of the dowel rotation.

REFLECTION

SURPRISES ENCOUNTERED

I thought that creating the staircase would be fairly straightforward because it only involves using cams and a dowel. However, my cams were incorrectly designed and my stairs were too heavy as well. I had to recreate the cams and stairs three times, preventing me from progressing in my project.

FRUSTRATING PARTS

As mentioned before, the cams were the most frustrating part because it seemed like the design and shape of my cams would work but they inevitably didn't. The motor was also difficult to work with since there was initially too little power so the dowel would stop rotating with any little friction. However, after removing the resistor in my circuit, there was now too much power so I have to design some gears to redistribute the speed.

DIFFERENT APPROACHES

I'm not done with the project yet so I'm not entirely sure on different approaches I would take but so far, I would make sure that my cams and staircases actually function properly before going through the hassle of printing all of the components out and gluing them together, all to find out that I have to restart. I would also make my stairs a bit lighter and with a smaller degree slope since my stairs don't resemble the conventional staircase which was an aspect I wanted.

THANK YOU!